Pure Mathematics with Mechanics GCE A Level 2024 - Paper 3

Duration: 3h | Cameroon GCE Board

General Certificate of Education Board - General Certificate of Education Examination. Pure Mathematics with Mechanics, Paper 3 (0765/3), ADVANCED LEVEL, June 2024. Time allowed: three hours.

Full marks may be obtained for answers to ALL questions. All questions carry equal marks. You are reminded of the necessity for good English and orderly presentation in your answers. Mathematical formulae booklet published by the Board are allowed. In calculations, you are advised to show all the steps in your working, giving your answer at each stage. Calculators are allowed. Start each question on a fresh page.

The scan supplied is complete: cover page and questions 1 to 8 (4 pages), eight questions all compulsory, each carrying 13 marks according to the marks printed in the margin.

Exercise 1

The position vector $\mathbf{r}$ of a particle at time t is given by $\mathbf{r} = [(2t+3)\mathbf{i} + (t^2-1)\mathbf{j}]$ m.

Find the

a. average velocity over the interval $t = 0$ to $t = 3$ s,

b. speed and distance of the particle from the origin when $t = 1$ s,

c. Cartesian equation of its path.

Exercise 2

A car travels on a straight horizontal track. The car decelerates uniformly from a speed of $20\ \mathrm{m\,s^{-1}}$ to a speed of $12\ \mathrm{m\,s^{-1}}$ as it travels a distance of 640 m. The car then accelerates uniformly, travelling a further 1820 m in 70 s. Calculate the

a. time that it takes the car to travel the first 640 m,

b. deceleration of the car during the first 640 m,

c. acceleration of the car as it travels the further 1820 m,

d. speed of the car when it has completed the further 1820 m,

e. average speed of the car as it travels the 2460 m.

Exercise 3

Two particles P and Q, of masses $6\ \text{kg}$ and $4\ \text{kg}$ respectively, are connected by a light inextensible string of length $2\ \text{m}$. The string passes over a light smooth pulley fixed at the top of a smooth plane which is inclined at an angle α to the horizontal, where $\sin\alpha = \frac{3}{5}$. Initially, the particles are held at rest with the string taut, P lying on the plane and Q hanging over the pulley. The particles are then released from rest.

(Take g as $10\ \text{m s}^{-2}$.)

a. Find the magnitude of the acceleration of the particles and the tension in the string.

b. Given that particle Q is initially $1.5\ \text{m}$ above the ground, and particle P does not reach the top of the plane, calculate the speed which particle Q hits the ground.

c. Under the conditions of part (b), calculate the time that elapses between when Q reaches the ground and when the string becomes taut again.

Exercise 4

Particles A and B, of masses $6\ \text{kg}$ and $2\ \text{kg}$ respectively, collide directly. The speeds of A and B after collision are $4\ \text{m s}^{-1}$ and $6\ \text{m s}^{-1}$ respectively and the coefficient of restitution between A and B is $\frac{1}{3}$.

a. Calculate the initial velocities of A and B given that they were moving in the same direction.

b. B moves and hits directly another stationary particle C of mass $m\ \text{kg}$, causing C to gain a speed of $10\ \text{m s}^{-1}$. Find m. [The printed question gives no coefficient of restitution for the impact between B and C and no speed for B after it: the model answer states the extra condition it uses.]

c. B moves and hits directly another stationary particle C of mass $m\ \text{kg}$, causing C to gain a speed of $10\ \text{m s}^{-1}$. Find the kinetic energy lost by the system. [Same missing condition as in (b): see the model answer.]

Exercise 5

i. A uniform ladder AB, of weight $40\ \text{N}$ and length $8\ \text{m}$, is resting with A against a smooth vertical wall and B on rough horizontal ground. The coefficient of friction between the ladder and the ground is $\frac{1}{4}$. The ladder makes an angle θ with the horizontal ground, where $\tan\theta = \frac{3}{4}$. Given that, towards the wall, a horizontal force of magnitude $125\ \text{N}$ is applied at a point D on the ladder, where $BD = 1.5\ \text{m}$, find how far up the ladder a man of weight $200\ \text{N}$ will climb before the ladder will begin to slip.

ii.a. An elastic string AB of natural length $1.5\ \text{m}$ is hanging from a fixed point A. A mass of $5\ \text{kg}$ is suspended from the end B and produces an extension of $3\ \text{mm}$. Calculate the modulus of elasticity of the string. (Take g as $10\ \text{m s}^{-2}$.)

ii.b. An elastic string AB of natural length $1.5\ \text{m}$ is hanging from a fixed point A. A mass of $5\ \text{kg}$ is suspended from the end B and produces an extension of $3\ \text{mm}$. Calculate the elastic potential energy of the string. (Take g as $10\ \text{m s}^{-2}$.)

Exercise 6

i.a. A car of mass $750\ \text{kg}$ is moving up a straight road inclined at an angle θ to the horizontal, where $\sin\theta = \frac{1}{15}$. The resistance to the motion of the car from non-gravitational forces has a constant magnitude of R newtons. The power developed

by the car's engine is $15\ \text{kW}$ and the car is moving at a constant speed of $20\ \text{m s}^{-1}$. Find R. (Take g as $10\ \text{m s}^{-2}$.)

i.b. A car of mass $750\ \text{kg}$ is moving up a straight road inclined at an angle θ to the horizontal, where $\sin\theta = \frac{1}{15}$. The resistance to the motion of the car from non-gravitational forces has a constant magnitude of R newtons. The power developed by the engine of the car is increased to $18\ \text{kW}$ and the non-gravitational resistance remains the same, when the car is moving up a road at a speed of $20\ \text{m s}^{-1}$. Find its acceleration. (Take g as $10\ \text{m s}^{-2}$.)

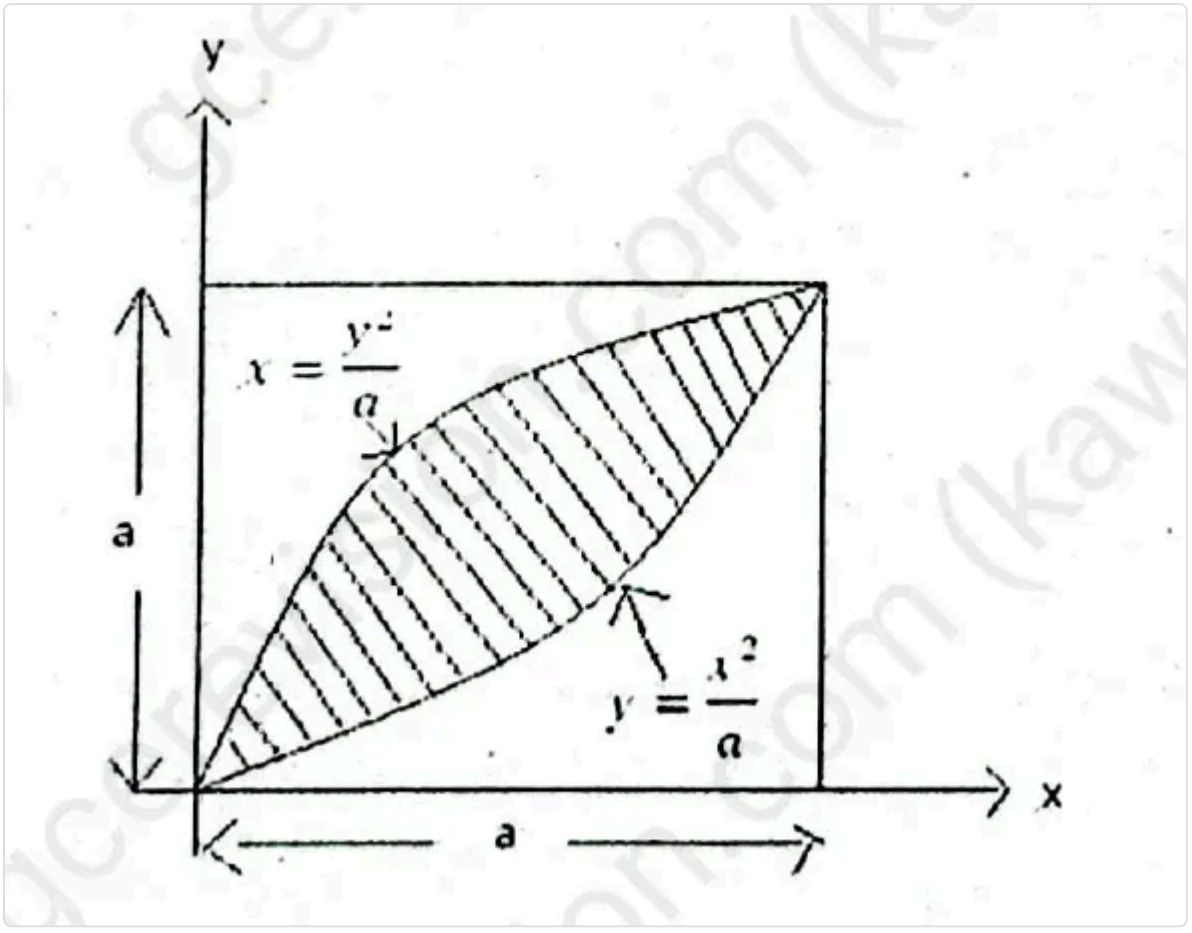
ii. A bead is threaded on a straight wire. The vector equation of the wire is $\mathbf{r} = \mathbf{i} - 3\mathbf{j} + \mathbf{k} + t(2\mathbf{i} - \mathbf{j} + 2\mathbf{k})$ m. The bead is moved from point A on the wire through a distance of 6 m along the wire to a point B by a force $\mathbf{F} = (7\mathbf{i} + 4\mathbf{j} - 2\mathbf{k})$ N. Find the magnitude of the work done by $\mathbf{F}$ in moving the bead from A to B.

Exercise 7

i.a. A trapezium $ABCD$ with uniform density is such that $AB = 4$ m and $BC = CD = DA = 2$ m. Show that the centre of mass of the lamina is $\frac{4\sqrt{3}}{9}$ m from AB.

i.b. A trapezium $ABCD$ with uniform density is such that $AB = 4$ m and $BC = CD = DA = 2$ m. The lamina is freely suspended from D and hangs in equilibrium. Find the angle between DC and the vertical through D.

ii.c. Given that the shaded area in the figure below is bounded by the two curves given by $x = \frac{y^2}{a}$ and $y = \frac{x^2}{a}$,



[Printed label (c).] Show, by integration, that the x-coordinate of the centroid of the shaded area is $\frac{9}{20}a$.

Exercise 8

i.a. Two events A and B are such that $P(A) = \frac{3}{10}$ and $P(A \cup B) = \frac{4}{5}$. Find $P(B)$ if A and B are mutually exclusive.

i.b. Two events A and B are such that $P(A) = \frac{3}{10}$ and $P(A \cup B) = \frac{4}{5}$. Find $P(B)$ if A and B are independent.

ii.c. Bag X contains 7 balls, of which 2 are red, 4 are green and 1 is blue. Bag Y contains 6 balls, of which 1 is red, 2 are green and 3 are blue. A biased die is thrown and the probability of obtaining an even score is $\frac{3}{5}$. If the score on the die is even, Bag X is chosen, otherwise Bag Y is chosen. A ball is selected at random from a bag. Draw a tree diagram showing all the possible outcomes.

ii.d. Bag X contains 7 balls, of which 2 are red, 4 are green and 1 is blue. Bag Y contains 6 balls, of which 1 is red, 2 are green and 3 are blue. A biased die is thrown and the probability of obtaining an even score is $\frac{3}{5}$. If the score on the die is even, Bag X is chosen, otherwise Bag Y is chosen. A ball is selected at random from a bag. Hence, find the probability that the ball is red.

ii.e. Bag X contains 7 balls, of which 2 are red, 4 are green and 1 is blue. Bag Y contains 6 balls, of which 1 is red, 2 are green and 3 are blue. A biased die is thrown and the probability of obtaining an even score is $\frac{3}{5}$. If the score on the die is even, Bag X is chosen, otherwise Bag Y is chosen. A ball is selected at random from a bag. Hence, find the probability that the ball is green.

ii.f. Bag X contains 7 balls, of which 2 are red, 4 are green and 1 is blue. Bag Y contains 6 balls, of which 1 is red, 2 are green and 3 are blue. A biased die is thrown and the probability of obtaining an even score is $\frac{3}{5}$. If the score on the die is even, Bag X is chosen, otherwise Bag Y is chosen. A ball is selected at random from a bag. Given that the ball is blue, find the probability that it is from Bag Y. [This last part has no printed label; numbered (f) here.]

Want to check your answers?

The full correction of this paper, with the method explained step by step, is available on EduKora: 15,000 FCFA per child until the exam, 12,000 FCFA for each additional child in the same family.

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